

→ Tokenization of all sentences

1. I love learning Artificial Intelligence

Tokens:

["I", "love", "learning", "Artificial", "intelligence"]

2. The cat is sleeping on the sofa.

Tokens:

["The", "cat", "is", "sleeping", "on", "the", "sofa"]

3. Students are studying in the library.

Tokens:

["Students", "are", "studying", "in", "the", "library"]

4. The weather is pleasant, today.

Tokens:

["The", "weather", "is", "pleasant", "today"]

5. My friend plays cricket every weekend.

Tokens:

["My", "friend", "plays", "cricket", "every", "weekend"]

6. She likes reading story books

Tokens:

["She", "likes", "reading", "story", "books"]

7. We are working on a machine learning project

Tokens:

['we', 'are', 'working', 'on', 'a', 'machine', 'learning', 'project']

8. The teacher explained the lesson clearly

['The', 'teacher', 'explained', 'the', 'lesson', 'clearly']

9. Birds are flying in the sky

Tokens:

['Birds', 'are', 'flying', 'in', 'the', 'sky']

10. They completed the assignment on time

Tokens

['They', 'completed', 'the', 'assignment', 'on', 'time']

→ Embedding concept

1. "I love learning Artificial Intelligence"

Tokens:

['I', 'love', 'learning', 'Artificial', 'Intelligence']

Embeddings

Embedding vectors

love

learning

Artificial

Intelligence

[0.12, 0.45, 0.79]

[0.65, 0.92, 0.11]

[0.23, 0.76, 0.91]

[0.89, 0.15, 0.34]

2. "The cat is sleeping on the sofa"

word

embedding vector

The

$[0.11, 0.31, 0.56]$

cat

$[0.92, 0.61, 0.21]$

is

$[0.19, 0.41, 0.79]$

sleeping

$[0.54, 0.99, 0.39]$

on

$[0.17, 0.26, 0.73]$

sofa

$[0.01, 0.41, 0.15]$

→ Analysis using RNN

1. "I love learning Artificial Intelligence"

• RNN reads one word at a time

I

→ step-1

↓

love

→ step-2

↓

learning

→ step-3

↓

Artificial

→ step-4

↓

Intelligence → step-5

Final understanding

I + love + learning + Artificial + Intelligence

= person like studying AI

→ Transformer process

It reads the entire sentence at once

[I] [love] [learning] [Artificial] [Intelligence]

2. "We are working on a machine learning project"

RNN reads words sequentially

We

↓

are

↓

working

↓

a

↓

machine

↓

learning

↓

project

Final understanding

we + are + working + on + a + machine + learning + project

→ Transformer process

[we] [are] [working] [on] [a] [machine] [learning]
[project]